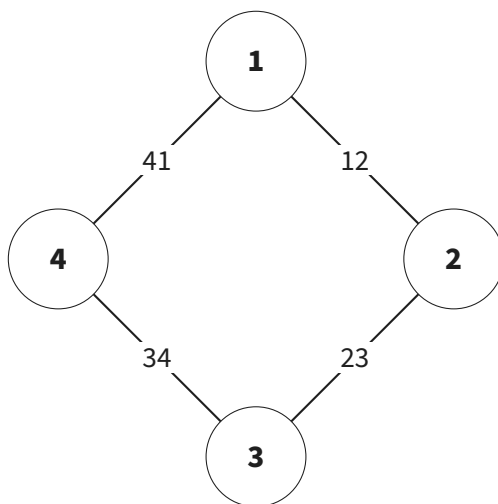


Problem 1: Put a two-sided counter on each lamp, all white (OFF). One move flips both ends of a drawn line; dark is ON. Try a move and its undo. Write 12 for the line from 1 to 2; write 12, 23 for those two moves in order.



Problem 2: Reset to all OFF for each target. Make only the listed lamps ON. Record a move list someone else can replay.

ON lamps	My moves
1 and 2	
1 and 3	
2 and 4	
1, 2, 3, 4	

Replay one list. Draw the picture it makes.

Problem 3: Use the four-lamp mat on page 1. Start all OFF for each target below. Try several move lists. Record a list that works, or mark “not yet” if none worked.

ON lamps	Moves, or not yet
1 only	
2 and 3	
1, 2, 3	
1 and 4	

Problem 4: Set the ends of line 12 to each starting picture below. Leave lamps 3 and 4 OFF. Press 12 once. Draw the two end lamps afterward, then count how many lamps are ON before and after.

Ends before	Ends after	ON before	ON after
			
			
			

Problem 5: Sort the targets you tried into those you made and those you have not made. Compare their numbers of ON lamps. Record a pattern to test with another target.

Problem 6: Start all OFF. Press the path 12, 23, 34. Draw the final lamps. Reset and press 14, 43. Draw that result too. On the mat, point to lamps flipped twice and lamps flipped once.

Final lamps for the two paths:

Problem 7: Make a collection of different reachable pictures. Use the mat, then record each picture as four circles in order 1, 2, 3, 4; fill an ON circle. For example, lamps 1 and 3 ON is ● ○ ● ○. Include all OFF. There may be extra boxes.

Problem 8: Sort your collection by the number of ON lamps. Choose a picture that has no move list yet and find one. Record your total and the new move list.